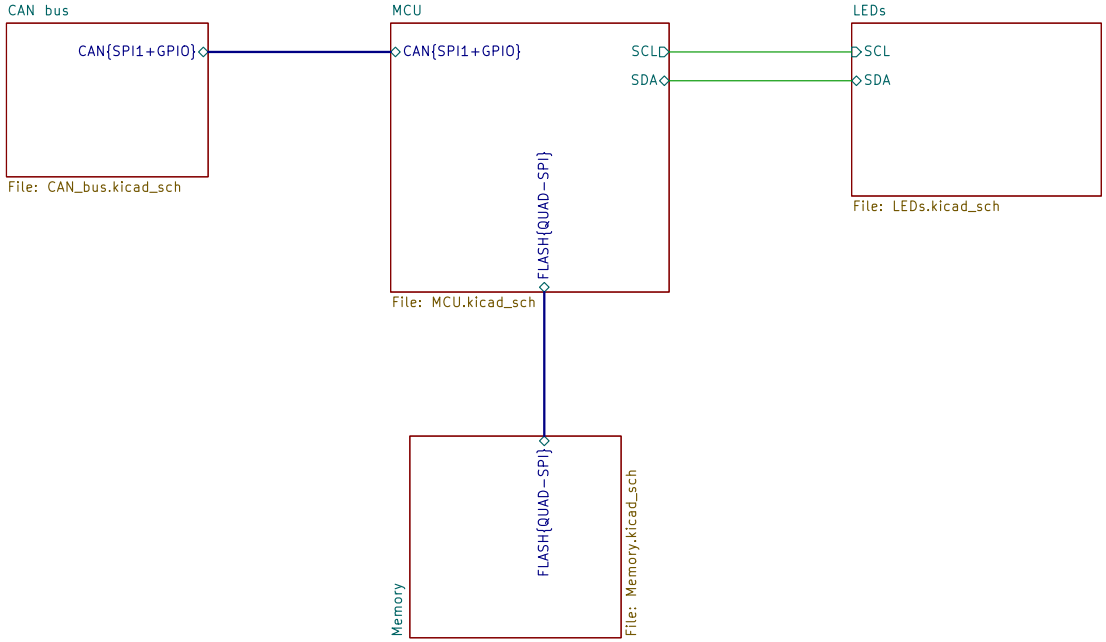


H101 MountingHole

H102 MountingHole

H103 MountingHole

H104 MountingHole

H105 MountingHole

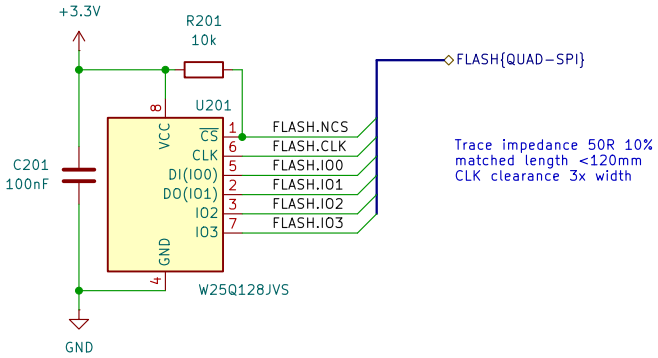
RnD Unit
rnd@icm-institute.org
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Sheet: /
File: BB_lighting_V1.kicad_sch

Title: BB_lighting

Size: A4	Date: 2024-03-21	**Rev: v1**
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Quad-SPI FLASH NOR Memory

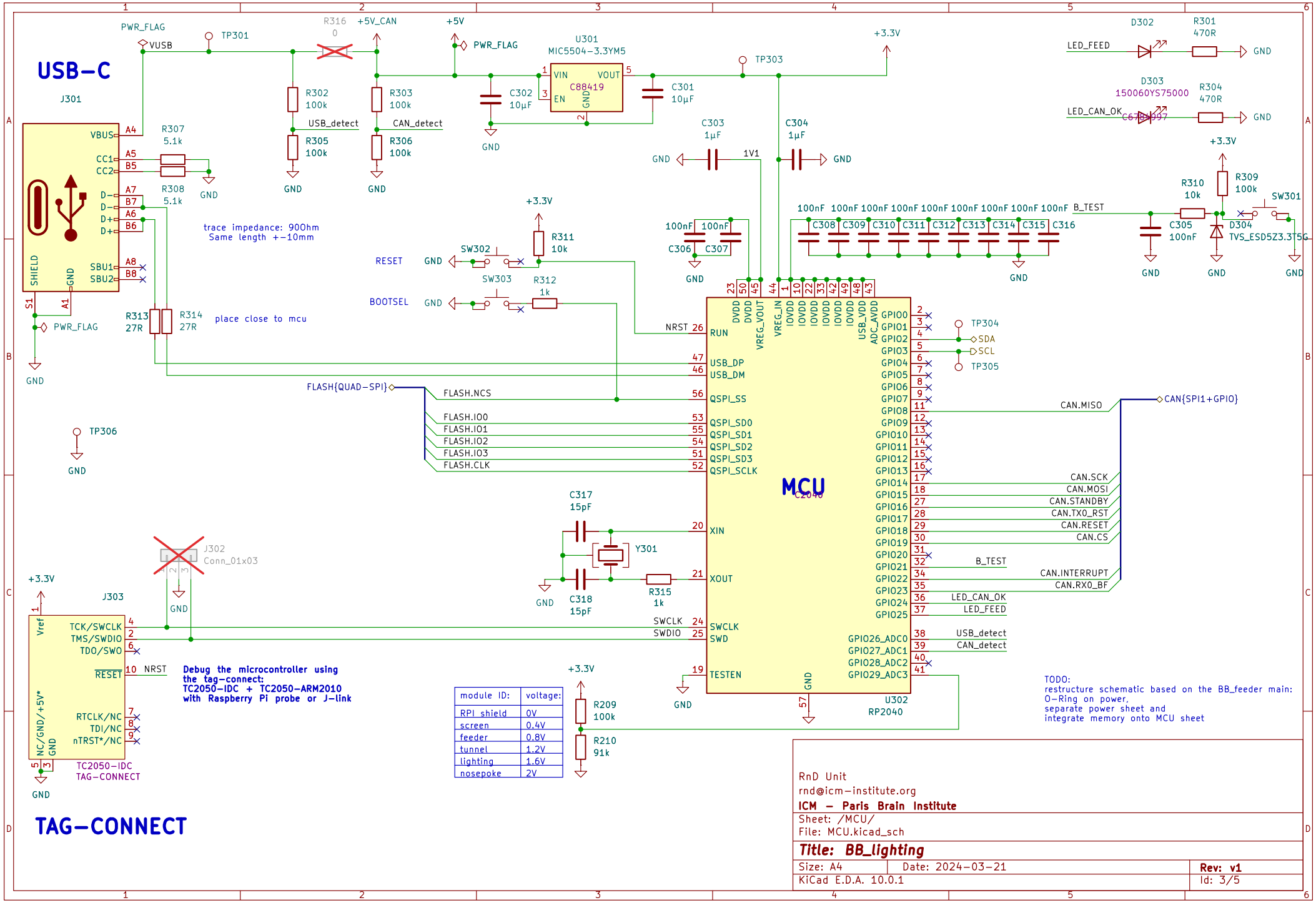


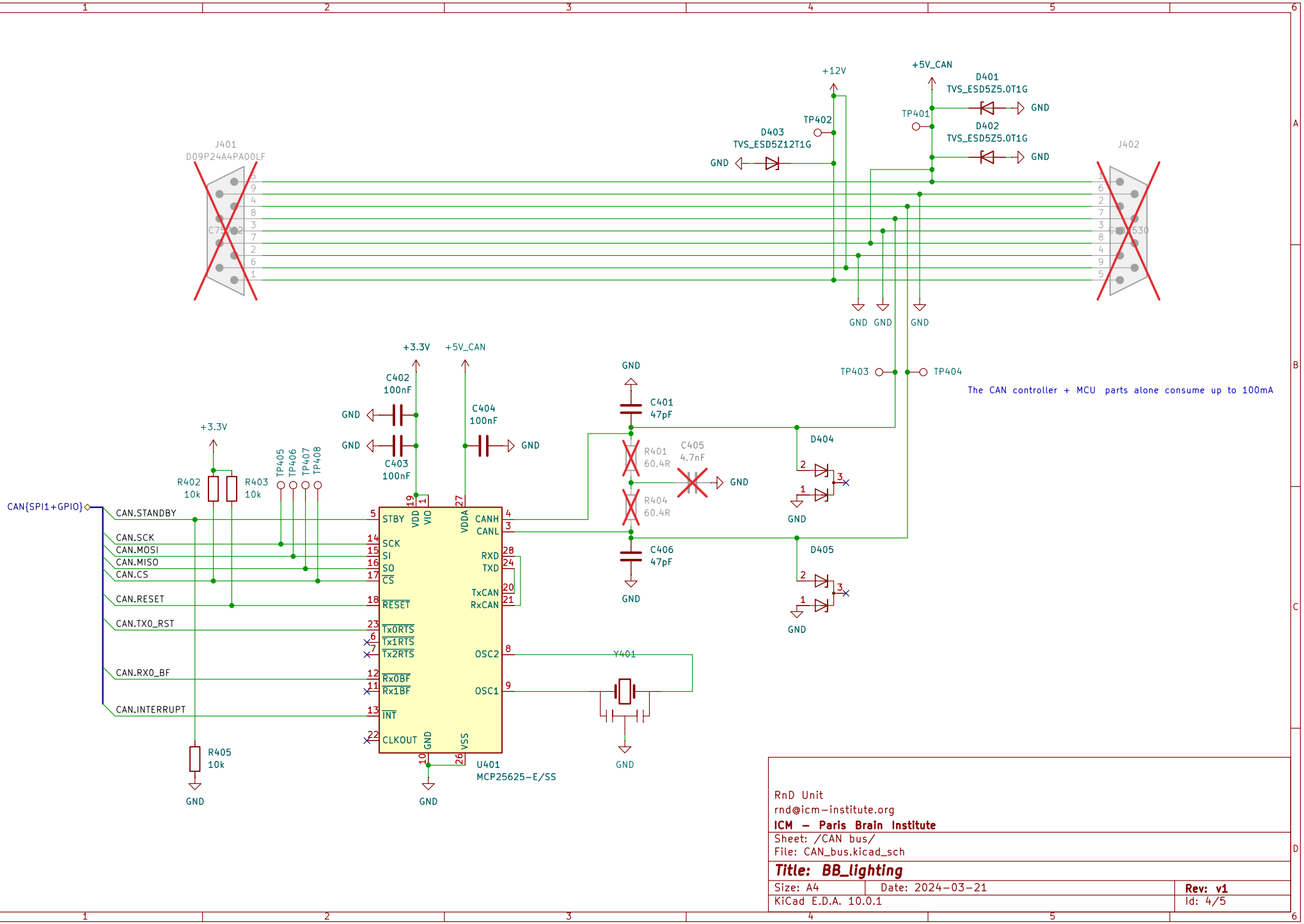
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File: Memory.kicad_sch

Title: BB_lighting

Size: A4 | Date: 2024-03-21 | **Rev: v1**
KiCad E.D.A. 10.0.1 | Id: 2/5


USB-C
J301
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
A4
A5
B5
A7
B7
A6
B6
A8
B8
S1
A1
PWR_FLAG
VUSB
TP301
R316
0
+5V_CAN
+5V
PWR_FLAG
U301
MIC5504-3.3YM5
C88419
VIN
VOUT
EN
GND
C302
10µF
C301
10µF
TP303
+3.3V
R302
100k
R303
100k
USB_detect
CAN_detect
R305
100k
R306
100k
R307
5.1k
R308
5.1k
trace impedance: 90Ohm
Same length +-10mm
C303
1µF
1V1
C304
1µF
D302
LED_FEED
R301
470R
D303
150060YS75000
LED_CAN_OK
R304
470R
R310
10k
R309
100k
SW301
B_TEST
D304
TVS_ESD5Z3.3T5G
C305
100nF
C306
C307
C308
C309
C310
C311
C312
C313
C314
C315
C316
100nF
SW302
RESET
SW303
BOOTSEL
R311
10k
R312
1k
R313
27R
R314
27R
place close to mcu
MCU
RP2040
U302
DVDD
VREG_VOUT
VREG_IN
IOVDD
USB_VDD
ADC_AVDD
RUN
USB_DP
USB_DM
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
XIN
XOUT
SWCLK
SWD
TESTEN
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
NRST
FLASH{QUAD-SPI}
FLASH.NCS
FLASH.IO0
FLASH.IO1
FLASH.IO2
FLASH.IO3
FLASH.CLK
TP304
SDA
SCL
TP305
CAN.MISO
CAN{SPI1+GPIO}
CAN.SCK
CAN.MOSI
CAN.STANDBY
CAN.TXO_RST
CAN.RESET
CAN.CS
CAN.INTERRUPT
CAN.RX0_BF
LED_CAN_OK
LED_FEED
USB_detect
CAN_detect
TP306
C317
15pF
C318
15pF
Y301
R315
1k
J302
Conn_01x03
J303
Vref
TCK/SWCLK
TMS/SWDIO
TDO/SWO
RESET
RTCLK/NC
TDI/NC
nTRST*/NC
NC/GND/+5V*
GND
TC2050-IDC
TAG-CONNECT
TAG-CONNECT
Debug the microcontroller using
the tag-connect:
TC2050-IDC + TC2050-ARM2010
with Raspberry Pi probe or J-link
R209
100k
R210
91k
module ID:
voltage:
RPI shield
0V
screen
0.4V
feeder
0.8V
tunnel
1.2V
lighting
1.6V
nosepoke
2V
TODO:
restructure schematic based on the BB_feeder main:
O-Ring on power,
separate power sheet and
integrate memory onto MCU sheet
RnD Unit
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Sheet: /MCU/
File: MCU.kicad_sch
Title: BB_lighting
Size: A4
Date: 2024-03-21
Rev: v1
KiCad E.D.A. 10.0.1
Id: 3/5

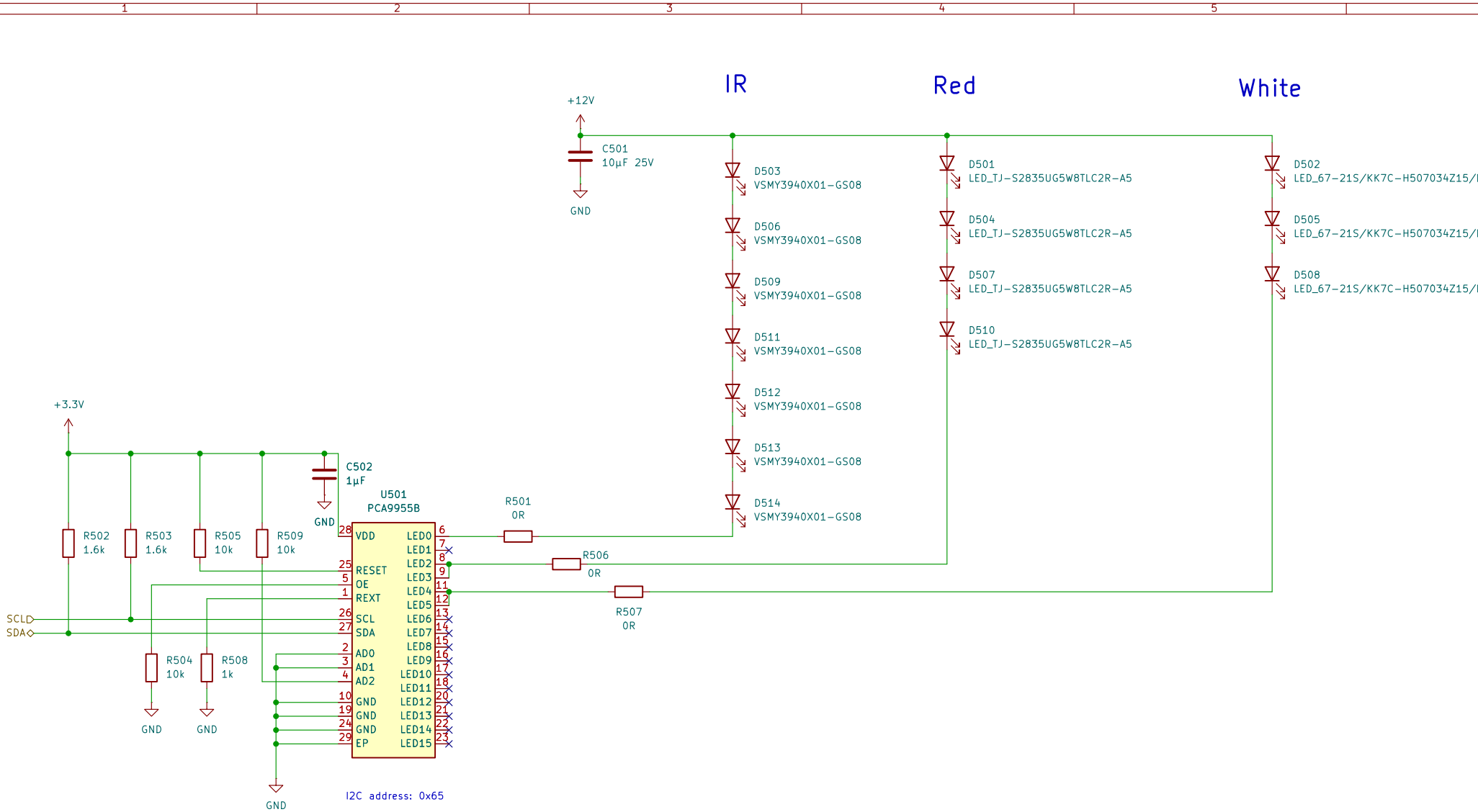


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Title: BB_lighting

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File: LEDs.kicad_sch	
Title: BB_lighting	
Size: A4	Date: 2024-03-21
KiCad E.D.A. 10.0.1	Id: 5/5
	Rev: v1